GLM_Gamma_InverseGaussian

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```
library(readxl)
library(Metrics) # RMSE/RMSLE
library(dplyr)
library(ggplot2)
library(dgof) #ks.test
library(fitdistrplus) #MLE
library(actuar)
library(SuppDists)
library(EnvStats)
```

Fokus utama pada GLM dengan tipe respons kontinu adalah memprediksi ekspektasi severity suatu kejadian. Pada collective risk model, ekspektasi severity kejadian ini dinyatakan sebagai E[X].

Gamma GLM

1) total: settled amount
2) inj1, ..., inj5: injury 1,..., injury 5 coded as 1 = no injury; 2, 3, 4, 5 = injury severities; 6 = fatal injury; 9 = not recorded
3) legrep: legal representation (0 = no, 1 = yes)
4) accmonth: accident month (1 = July 1989, ..., 120 = June 1999)
5) repmonth: reporting month (as above)
6) finmonth: finalization month (as above)
7) op_time: operational time

```
data=persinj <- read_excel("persinj.xls")

## Warning: Expecting numeric in D8128 / R8128C4: got 'N'

data$legrep=as.factor(data$legrep)
head(data)

## # A tibble: 6 x 11
##    total  inj1  inj2  inj3  inj4  inj5 legrep accmonth repmonth finmonth
op_time
##    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <fct>     <dbl>    <dbl>    <dbl>
```

```
<dbl>
## 1   87.7     1     0     0     0     0 0            50        51        52
0.1
## 2  354.      1     0     0     0     0 0            54        55        56
0.1
## 3  689.      1     0     0     0     0 0            55        55        56
0.1
## 4  173.      1     0     0     0     0 0            58        58        59
0.1
## 5   43.3     1     0     0     0     0 0            62        63        63
0.1
## 6 2915.      6     0     0     0     0 0            66        67        67
0.1
```

```
str(data)
```

```
## tibble [22,036 x 11] (S3: tbl_df/tbl/data.frame)
##  $ total   : num [1:22036] 87.7 353.6 688.8 172.8 43.3 ...
##  $ inj1    : num [1:22036] 1 1 1 1 1 6 1 1 1 1 ...
##  $ inj2    : num [1:22036] 0 0 0 0 0 0 2 1 1 0 ...
##  $ inj3    : num [1:22036] 0 0 0 0 0 0 1 1 0 0 ...
##  $ inj4    : num [1:22036] 0 0 0 0 0 0 0 2 0 0 ...
##  $ inj5    : num [1:22036] 0 0 0 0 0 0 0 0 0 0 ...
##  $ legrep  : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
##  $ accmonth: num [1:22036] 50 54 55 58 62 66 68 71 74 77 ...
##  $ repmonth: num [1:22036] 51 55 55 58 63 67 68 71 74 77 ...
##  $ finmonth: num [1:22036] 52 56 56 59 63 67 69 73 75 78 ...
##  $ op_time : num [1:22036] 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 ...
```

```
summary(data)
```

```
##      total              inj1          inj2              inj3
##  Min.   :     10   Min.   :1.00   Min.   :0.0000   Min.   :0.0000
##  1st Qu.:   6297   1st Qu.:1.00   1st Qu.:0.0000   1st Qu.:0.0000
##  Median :  13854   Median :1.00   Median :1.0000   Median :0.0000
##  Mean   :  38367   Mean   :1.83   Mean   :0.6654   Mean   :0.4169
##  3rd Qu.:  35123   3rd Qu.:2.00   3rd Qu.:1.0000   3rd Qu.:1.0000
##  Max.   :4485797   Max.   :9.00   Max.   :9.0000   Max.   :9.0000
##                                                    NA's   :1
##       inj4              inj5          legrep      accmonth          repmonth
##  Min.   :0.0000   Min.   :0.0000   0: 8008   Min.   :  1.00   Min.   :
15.00
##  1st Qu.:0.0000   1st Qu.:0.0000   1:14028   1st Qu.: 44.00   1st Qu.:
54.00
##  Median :0.0000   Median :0.0000             Median : 64.00   Median :
69.00
##  Mean   :0.2405   Mean   :0.1575             Mean   : 62.05   Mean   :
71.33
##  3rd Qu.:0.0000   3rd Qu.:0.0000             3rd Qu.: 82.00   3rd Qu.:
85.00
##  Max.   :9.0000   Max.   :9.0000             Max.   :115.00   Max.
```

```
:116.00
##
##     finmonth         op_time
##  Min.   : 49.00   Min.   : 0.10
##  1st Qu.: 72.00   1st Qu.:23.00
##  Median : 91.00   Median :45.90
##  Mean   : 88.37   Mean   :46.33
##  3rd Qu.:106.00   3rd Qu.:69.30
##  Max.   :117.00   Max.   :99.10
##
```

Cek Distribusi Respons

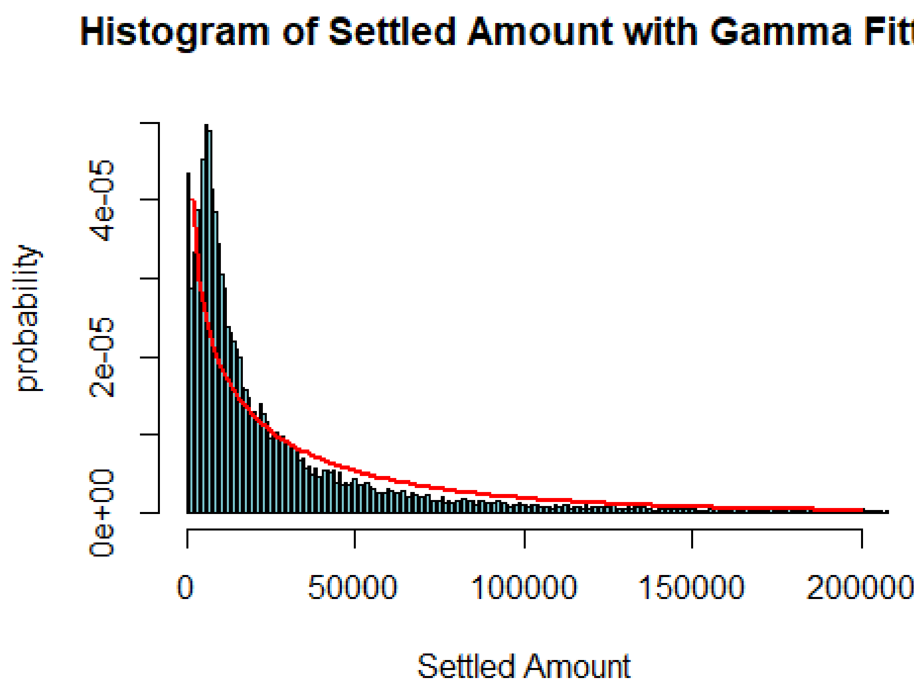
Lakukan MLE Gamma kepada Settled Amount of claim

```
Gam=summary(fitdist(data$total/1000000,"gamma"))
Gam$estimate

##      shape       rate
##  0.6162608 16.0598252

meandata=Gam$estimate[1]*(1/Gam$estimate[2])*1000000;meandata

##    shape
## 38372.82

variansidata=(Gam$estimate[1])*((1/Gam$estimate[2])^2)*(1000000^2);variansida
ta

##      shape
## 2389367245

scalebintang=variansidata/(meandata);scalebintang

##    shape
## 62267.18

shapebintang=meandata/(scalebintang);shapebintang

##     shape
## 0.6162608
```

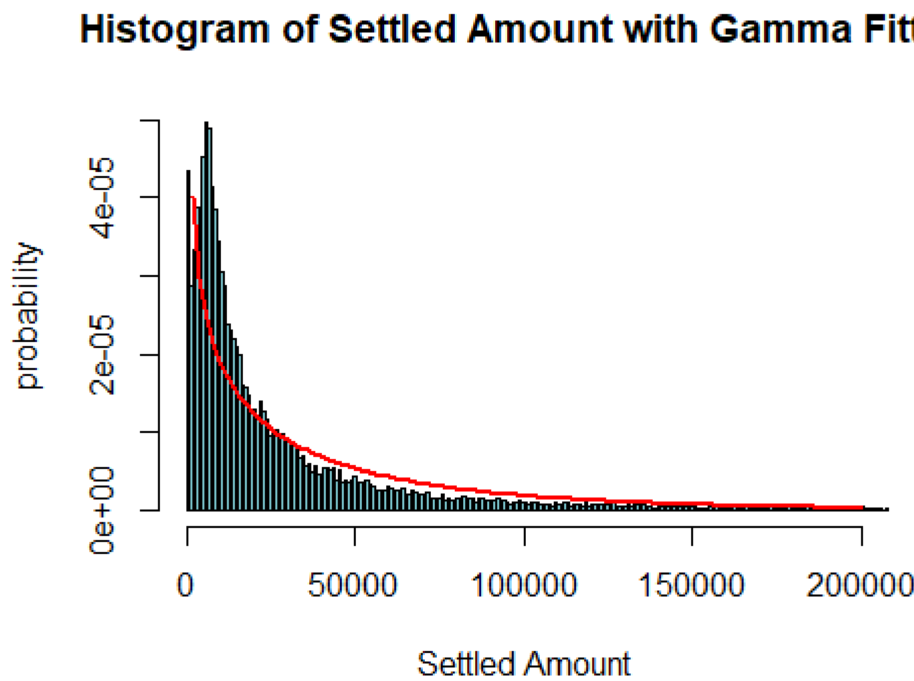
Pertama-tama, lihat dahulu secara grafis apakah data ini cocok dengan distribusi gamma

```
#dengan parameter shape & scale
h = hist(data$total,xlim=c(0,200000),
         breaks = seq(0,4500000,900),probability = T,
         main = paste("Histogram of Settled Amount with Gamma Fitted"),
         col="cadetblue3",
         xlab = "Settled Amount",
         ylab = "probability")
```

```r
curve(dgamma(x,
                shape = 0.6162608   ,
                scale=1000000/16.0598252),
                add=TRUE,
                lwd=2,
                col="red")
```



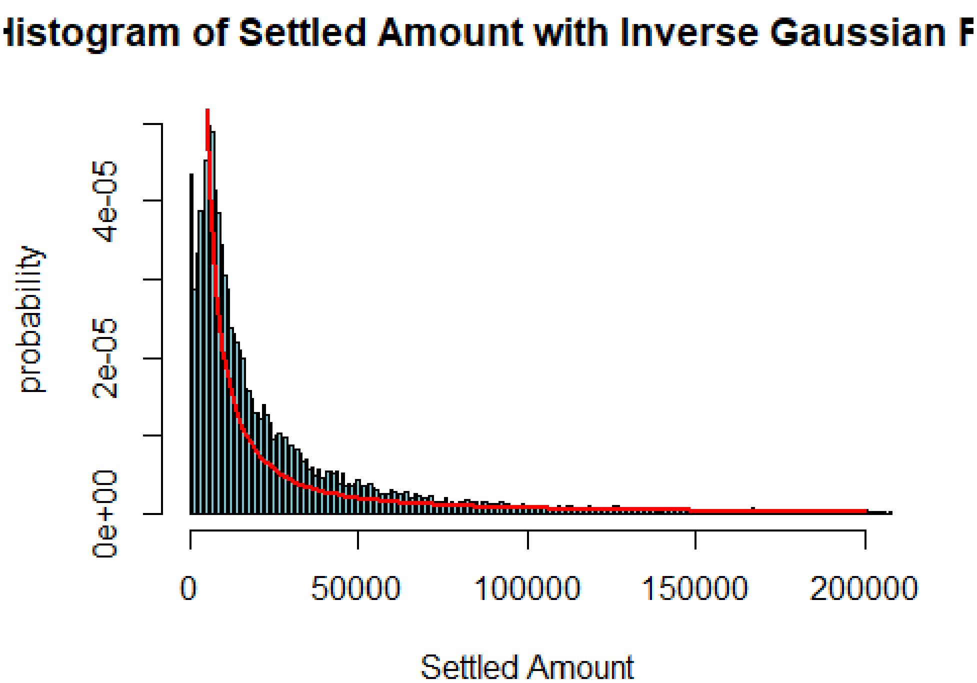

```r
#dengan parameter shape & rate
h = hist(data$total,xlim=c(0,200000),
         breaks = seq(0,4500000,900),probability = T,
         main = paste("Histogram of Settled Amount with Gamma Fitted"),
         col="cadetblue3",
         xlab = "Settled Amount",
         ylab = "probability")
curve(dgamma(x,
                shape = 0.6162608   ,
                rate = 16.0598252   /1000000),
                add=TRUE,
                lwd=2,
                col="red")
```



Lakukan hal yang serupa dengan inverse Gaussian

```
Gau=summary(fitdist(data$total/1000000,"invgauss",lower=c(0,0)
                    ,start = list(mean = 0.5, shape = 5)))

## Warning in checkparamlist(arg_startfix$start.arg, arg_startfix$fix.arg, :
Some
## parameter names have no starting/fixed value but have a default value:
## dispersion.

Gau$estimate

##        mean       shape
## 0.038371236 0.003249311

meandata=Gau$estimate[1]*1000000;meandata

##     mean
## 38371.24

variansi=((Gau$estimate[1]^3)/Gau$estimate[2])*(1000000^2);variansi

##        mean
## 17387059270

lambdadata = meandata^3/variansi ; lambdadata

##     mean
## 3249.311
```

```
h = hist(data$total,xlim=c(0,200000),
         breaks = seq(0,4500000,900),probability = T,
         main = paste("Histogram of Settled Amount with Inverse Gaussian
Fitted"),
         col="cadetblue3",
         xlab = "Settled Amount",
         ylab = "probability")
curve(dinvGauss(x,
                nu = 0.038371236*1000000, #Mean
                lambda = lambdadata  ), #Shape
                add=TRUE,
                lwd=2,
                col="red")
```




```
Gam$aic

## [1] -103808

Gau$aic

## [1] -92776.09

Gam$bic

## [1] -103792

Gau$bic
```

```
## [1] -92760.09
```

Terlihat bahwa distribusi Gamma memiliki nilai AIC yang lebih kecil daripada inverse gaussian, maka distribusi yang lebih cocok untuk respon Settled Amount ini adalah distribusi Gamma.

Lakukan Uji Kolmogorov-Smirnov untuk memastikan

```
nrow(data)

## [1] 22036

set.seed(44)
ks.test(data$total,
        rgamma(22036,
               shape = 0.6162608,
               rate = 16.0598252/1000000),
        alternative = "two.sided",
        exact = TRUE)

##
##  Two-sample Kolmogorov-Smirnov test
##
## data:  data$total and rgamma(22036, shape = 0.6162608, rate =
16.0598252/1e+06)
## D = 0.11322, p-value < 2.2e-16
## alternative hypothesis: two-sided
```

Kesimpulannya, data juga tidak berdistribusi gamma karena p-value yang dihasilkan dari uji Kolmogorov-Smirnov lebih kecil daripada taraf signifikansi.

Karena ini hanya ilustrasi, kami berikan contoh membuat model GLM nya, meskipun data tidak berdistribusi gamma (dan inverse gaussian)

Model GLM

Langsung dilakukan stepwise regression, ternyata variabel yang tidak dipakai hanya accident month

```
model1=step(glm(total ~ op_time+
                  legrep+
                  op_time*legrep+
                  inj1+
                  inj2+
                  inj3+
                  inj4+
                  inj5+
                  accmonth+
                  repmonth+
                  finmonth
```

```
              ,family = Gamma(link = "log")
              ,data=data)
              ,direction = "both"
              ,trace = FALSE)
summary(model1)
```

```
##
## Call:
## glm(formula = total ~ op_time + legrep + inj1 + inj2 + inj3 +
##     inj4 + inj5 + repmonth + finmonth + op_time:legrep, family =
Gamma(link = "log"),
##     data = data)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -3.4196  -0.9175  -0.3864   0.1592  10.6411
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)     9.0761876  0.0594672 152.625  < 2e-16 ***
## op_time         0.0279349  0.0007707  36.248  < 2e-16 ***
## legrep1         0.5616747  0.0417962  13.438  < 2e-16 ***
## inj1            0.0345484  0.0052808   6.542 6.19e-11 ***
## inj2            0.1994465  0.0125193  15.931  < 2e-16 ***
## inj3            0.0816175  0.0134815   6.054 1.44e-09 ***
## inj4            0.1154826  0.0161235   7.162 8.18e-13 ***
## inj5            0.0953100  0.0150929   6.315 2.76e-10 ***
## repmonth       -0.0274098  0.0013984 -19.601  < 2e-16 ***
## finmonth        0.0135472  0.0011600  11.678  < 2e-16 ***
## op_time:legrep1 -0.0055806  0.0007784  -7.169 7.77e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 2.234293)
##
##     Null deviance: 44008  on 22034  degrees of freedom
## Residual deviance: 22807  on 22024  degrees of freedom
##   (1 observation deleted due to missingness)
## AIC: 488152
##
## Number of Fisher Scoring iterations: 9
```

Bandingkan dengan Model di Buku

```
model2=step(glm(total ~ op_time+
                  legrep+
                  op_time*legrep
              ,family = Gamma(link = "log")
              ,data=data)
              ,direction = "both"
```

```
              ,trace = FALSE)
summary(model2)

## 
## Call:
## glm(formula = total ~ op_time + legrep + op_time * legrep, family = 
Gamma(link = "log"), 
##     data = data)
## 
## Deviance Residuals: 
##     Min       1Q   Median       3Q      Max  
## -3.6253  -0.9860  -0.4332   0.1345   9.9012  
## 
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)    
## (Intercept)      8.2118446  0.0329095 249.528  < 2e-16 ***
## op_time          0.0383149  0.0006311  60.707  < 2e-16 ***
## legrep1          0.4667864  0.0424613  10.993  < 2e-16 ***
## op_time:legrep1 -0.0049978  0.0008002  -6.246 4.29e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## (Dispersion parameter for Gamma family taken to be 2.432031)
## 
##     Null deviance: 44010  on 22035  degrees of freedom
## Residual deviance: 25412  on 22032  degrees of freedom
## AIC: 490944
## 
## Number of Fisher Scoring iterations: 6
```

Selang Kepercayaan

1) SK Koefisien

```
confint(model1, level=0.95)

##                        2.5 %       97.5 %
## (Intercept)      8.954063635  9.198844084
## op_time          0.026317749  0.029556532
## legrep1          0.477319173  0.645735126
## inj1             0.022280196  0.047004667
## inj2             0.170910766  0.228649360
## inj3             0.050692303  0.113973849
## inj4             0.076630383  0.156400716
## inj5             0.059648836  0.133315063
## repmonth        -0.030302352 -0.024516579
## finmonth         0.011188697  0.015904542
## op_time:legrep1 -0.007170483 -0.003991443

confint(model2, level=0.95)
```

```
##                      2.5 %       97.5 %
## (Intercept)      8.144976592  8.279560005
## op_time          0.037005730  0.039627295
## legrep1          0.381222215  0.552046848
## op_time:legrep1 -0.006626246 -0.003371724
```

2) SK Mean—disesuaikan dengan link functionnya (log)

```
head(exp(data.frame(predict(model1,
                      newdata = data,
                      interval = 'confidence',
                      level=0.95))) %>% mutate(SettledAmount= data$total))

##         fit      lwr      upr SettledAmount
## 1 4537.272 4226.709 4870.653       87.7455
## 2 4292.526 4007.109 4598.272      353.6201
## 3 4292.526 4007.109 4598.272      688.8309
## 4 4117.669 3849.073 4405.009      172.7955
## 5 3790.231 3549.907 4046.824       43.2922
## 6 4261.914 3953.622 4594.246     2915.4320

head(exp(data.frame(predict(model2,
                      newdata = data,
                      interval = 'confidence',
                      level=0.95))) %>% mutate(SettledAmount= data$total))

##         fit      lwr      upr SettledAmount
## 1 3698.476 3467.801 3944.495       87.7455
## 2 3698.476 3467.801 3944.495      353.6201
## 3 3698.476 3467.801 3944.495      688.8309
## 4 3698.476 3467.801 3944.495      172.7955
## 5 3698.476 3467.801 3944.495       43.2922
## 6 3698.476 3467.801 3944.495     2915.4320
```

Pengurangan Residual

```
anova(model1, test="Chisq")

## Analysis of Deviance Table
##
## Model: Gamma, link: log
##
## Response: total
##
## Terms added sequentially (first to last)
##
##
##                Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
## NULL                           22034      44008
## op_time         1  18234.2     22033      25774 < 2.2e-16 ***
## legrep          1    280.0     22032      25494 < 2.2e-16 ***
```

```
## inj1           1     27.4     22031      25467 0.0004591 ***
## inj2           1   1206.5     22030      24260 < 2.2e-16 ***
## inj3           1    169.4     22029      24091 < 2.2e-16 ***
## inj4           1    129.0     22028      23962 3.004e-14 ***
## inj5           1     52.4     22027      23910 1.289e-06 ***
## repmonth       1    687.8     22026      23222 < 2.2e-16 ***
## finmonth       1    308.6     22025      22913 < 2.2e-16 ***
## op_time:legrep 1    105.9     22024      22807 5.824e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model2, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model: Gamma, link: log
##
## Response: total
##
## Terms added sequentially (first to last)
##
##
##                Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
## NULL                           22035      44010
## op_time        1  18228.9     22034      25781 < 2.2e-16 ***
## legrep         1    280.7     22033      25500 < 2.2e-16 ***
## op_time:legrep 1     88.3     22032      25412 1.678e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#Prediksi

```
pred=exp(predict(model1,data))
pred2=exp(predict(model2,data))
head(data.frame(pred,pred2,data$total))
```

```
##       pred    pred2 data.total
## 1 4537.272 3698.476    87.7455
## 2 4292.526 3698.476   353.6201
## 3 4292.526 3698.476   688.8309
## 4 4117.669 3698.476   172.7955
## 5 3790.231 3698.476    43.2922
## 6 4261.914 3698.476  2915.4320
```

Hasil dari Prediksi untuk Settled Amount of Claim dapat dianalogikan sebagai E[X] pada collective risk model.